Yunyue Su

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Objective

My research focuses on self-improving AI agents under scarce and ill-defined evaluation, with healthcare as a primary application domain. I am particularly interested in how agents can autonomously evaluate their own progress, learn from limited or uncertain feedback, and continually improve when reliable external evaluation is costly, sparse, or difficult to define. During my doctoral studies, I aim to develop generalizable methods for autonomous self-improvement in such settings and investigate their application to complex healthcare tasks, including mental health counseling.

Education

Saw Swee Hock School of Public Health, National University of Singapore — *Aug. 2026 - Current*
PhD, Public Health; Advised by Associate Professor Mengling 'Mornin' Feng — Singapore

Computer Network Information Center, University of Chinese Academy of Sciences — *Sept. 2020 - Jun. 2023*
M.S., Computer Technology; Advised by Associate Professor Yuanping Chen and Xiaoqian Zhu — Beijing, China

College of Mathematics and Computer Science, Fuzhou University — *Sept. 2016 - Jun. 2020*
B.S., Computer Science and Technology; Advised by Professors Xiangwen Liao and Qingyuan Bai — Fujian, China

Publications and Patents

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[C.1] Yunyue Su, Jiahui Chen, Zao Jiang, Zhenyi Zhong, Liang Wang, Qiang Liu, Zhaoxiang Zhang. (2026). **SpectraLLM: Uncovering the Ability of LLMs for Molecular Structure Elucidation from Multi-Spectral Data**. [arXiv preprint]. arXiv:2508.08441. [Accepted for publication]. *International Conference on Learning Representations (ICLR)*.

[C.2] Bowen Fang, Wen Ye, Yunyue Su, Jinghao Zhang, Qiang Liu, Yesheng Liu, Xin Sun, Shu Wu, Jiabing Yang, Baole Wei, Liang Wang. (2026). **ToolWeaver: Weaving Collaborative Semantics for Scalable Tool Use in Large Language Models**. [Accepted for publication]. *International Conference on Learning Representations (ICLR)*.

[C.3] Yunyue Su, Zhang Jinshuai, Bowen Fang, Wen Ye, Jinghao Zhang, Bowen Song, Weiqiang Wang, Qiang Liu, Liang Wang. (2025). **Toolscaler: Scalable Generative Tool Calling via Structure-Aware Semantic Tokenization**. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 556-578. Association for Computational Linguistics. 2025, Suzhou, China. DOI: 10.18653/v1/2025.findings-emnlp.30.

[C.4] Wen Ye, Zhaocheng Liu, Gui Yuwei, Tingyu Yuan, Yunyue Su, Bowen Fang, Chaoyang Zhao, Qiang Liu, Liang Wang. (2025). **GenPilot: A Multi-Agent System for Test-Time Prompt Optimization in Image Generation**. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 929–958. Association for Computational Linguistics. 2025, Suzhou, China. DOI: 10.18653/v1/2025.findings-emnlp.49.

[C.5] Ge Shi*, Yunyue Su*, Yongliang Ma, and Ming Zhou. (2023). **A Hybrid Detection and Generation Framework with Separate Encoders for Event Extraction.** In *the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pp. 3163-3180. Association for Computational Linguistics. 2023, Dubrovnik, Croatia. DOI: 10.18653/v1/2023.eacl-main.231.

[J.1] Shu Wu, Zekun Li, Yunyue Su, Zeyu Cui, Xiaoyu Zhang, Liang Wang. (2025). **GraphFM: Graph Factorization Machines for Feature Interaction Modelling.** *Machine Intelligence Research*, 22(2), pp. 239-253. DOI: 10.1007/s11633-024-1505-5.

[P.1] Yongliang Ma, Yunyue Su, Ge Shi, and Ming Zhou. (2022). **Event extraction method, system and computer readable storage medium**. Lanzhou Technology Co ltd, Patent No. CN202210160088.8A. Registration Date: 2022-02-22, Grant Date: 2022-06-21, Publication Date: 2022-06-21.

[T.1] Yunyue Su. (2023). **Research on key technologies of document-level event extraction in financial risk Management field**. *Library and Information Center, Chinese Academy of Sciences*. Call number: LW212474.

* Equal Contribution.

EXPERIENCE

NLPR, Institute of Automation, CAS 2023 - 2026
Advised by Professor Qiang Liu, Shu Wu, and Liang Wang. Beijing, China
Lead researcher on national key projects; core team contributor and co-author of academic publications.

Researcher

- Co-authored the paper *SpectraLLM* with Prof. Qiang Liu. This paper proposes a unified language-based framework for molecular structure elucidation from diverse spectroscopic data, unifying multi-spectral signal alignment, structural reasoning, and SMILES generation within a single end-to-end generative process without modality-specific encoders or handcrafted rules.
- Co-authored the paper Toolscaler with Prof. Qiang Liu. This paper research efficient methods for enabling LLMs to invoke large-scale toolkits (over 47,000). Developed a structure-aware semantic tokenization scheme and a post-guided iterative training strategy, unifying tool representation, retrieval, and calling within a single generative process. [Code]
- Co-authored the paper *GraphFM* with Prof. Shu Wu. This work pioneered the integration of graph structures into Factorization Machines (FM) frameworks, modeling feature interactions from a topological perspective. By leveraging GNN-based aggregation strategies, it explicitly captures high-order interactions while automatically filtering noise, addressing inherent limitations of traditional FMs. [Code]

Intermediate Engineer

- **Cross-Modal Scientific Data Framework**: Developed a foundation large language model integrating different spectral to infer molecular structures and properties, which accuracy rate of molecular structure prediction has increased by 31% compared to existing algorithms. This framework has now been integrated into the **ScienceOne** platform and provides services for dozens of research institutions under the Chinese Academy of Sciences.[Website]
- **Algorithm Development Leadership**: Led R&D for industry-academia projects, including a national R&D sub-project on graph neural network-based industrial risk prediction (HuaXia Bank, 30 million CNY), Ant Group's retrieval-augmented controllable generation project (350K CNY), and Tencent's knowledge-driven LLM editing system (200K CNY).
- **RAG-based Multimodal Fact-Checking Platform**: Built a platform supporting multi-modal evidence retrieval and reasoning for offline/online fact-checking, deployed for clients such as the Chinese Academy of Sciences.

Computer Network Information Center, CAS 2020 - 2023
Graduate Student Researcher, Advised by Associate Professor Yuanping Chen and Xiaoqian Zhu.

- Designed a *document-level generative event extraction framework for financial risk management*, integrating domain knowledge-enhanced multi-task training to address cross-sentence argument extraction and multi-event argument sharing in low-resource long-text scenarios, outperforms previous SOTA with 1.1% avg. F1 gain. [Report]

Baidu, Inc. 2022 - 2023
Research Intern, Mentored by Tianfu Zhang.

- Developed a *probe-based* method to automatically generate syntactic dependency trees by extracting structured features from language models, enabling high-quality annotation for new texts and enhancing long-distance argument association modeling and extraction accuracy in event processing.

Langboat Co., Ltd 2021 - 2022
Research Intern, Mentored by Yongliang Ma, Assistant Professor Ge Shi, and Professor Ming Zhou.

- Developed a *generative event extraction framework* for bank announcement data, integrating constrained decoding and adaptive template mechanisms (soft prompt/dynamic prefix) to resolve error-prone template generation and few-shot adaptation challenges. Achieved above 10% accuracy improvement in argument extraction and deployed in production with a patent application (CN*) filed.
- Designed a separate-encoder framework with joint bridging training to mitigate feature interference between event detection and argument extraction, resulting in 5.7% and 2% improvement, respectively. Research published as a long paper in EACL 2023. [Code]

Ruijie Networks Co., Ltd 2019 - 2020
Audio processing intern, Mentored by Honghe Wu

- Responsible for collecting and labeling audio data for training speech separation models; re-engineered the popular audio processing framework Kaldi using Python to achieve operator acceleration.
- Implemented an OCR recognition algorithm for PDF documents, using multimodal similarity calculation algorithms to match recognized document blocks and enable document plagiarism detection, deployed online via Docker.

Supercomputing Center, Department of Computer Science, FuZhou University 2016 - 2020
Undergraduate Student Researcher, Advised by Professor Xiangwen Liao and Qingyuan Bai.

- Designed and implemented an *end-to-end music recommendation* algorithm by leveraging temporal convolution networks to extract deep features from audio Mel-spectrograms, integrating user behavior sparse features and audio semantic features via DeepFM model for user interest prediction, and generating personalized recommendations based on ranking.
- Under the supervision of Prof. Dong Zhang, collaborated with a team of 9 peers to design and develop Jarvis for Chat, an intelligent conversational assistant powered by large language models, as part of a software engineering capstone project. [Blogs, Github]